Class 9 Mathematics — Exercise 3.1

Basic Concept

(1) Venn Diagram

A Venn diagram shows sets as closed figures (circles) inside a rectangle. The rectangle represents the **universal set** U.

- **Disjoint sets:** sets with no common element, e.g. $A = \{1, 2, 3\}$, $B = \{4, 5\}$ — the circles do not touch.
- **Overlapping sets:** sets with some common elements, e.g. $A = \{2, 3, 5, 7, 9\}$, $B = \{0, 3, 5, 8\}$ — the circles overlap; the common part is $A \cap B = \{3, 5\}$.

(2) Shading Method

- **Union ($\cup$):** shade **all** the regions of both sets.
- **Intersection ($\cap$):** shade only the **common** region.
- First shade the part in brackets (shown with horizontal lines), then shade the final answer (shown with slanting lines).

(3) Laws of Sets

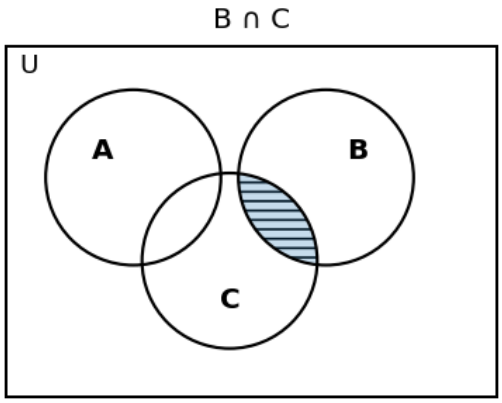
- **Associative law:** $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$
- **Distributive law of $\cup$ over $\cap$:** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- **Distributive law of $\cap$ over $\cup$:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

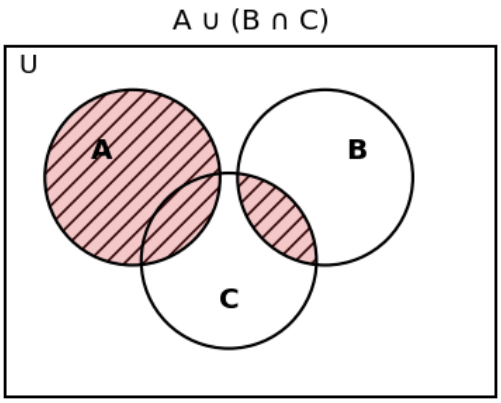
Question 1

Shade $A \cup (B \cap C)$, $A \cap (B \cup C)$, $(A \cup B) \cup C$ and $A \cap (B \cap C)$ using the following Venn diagrams.

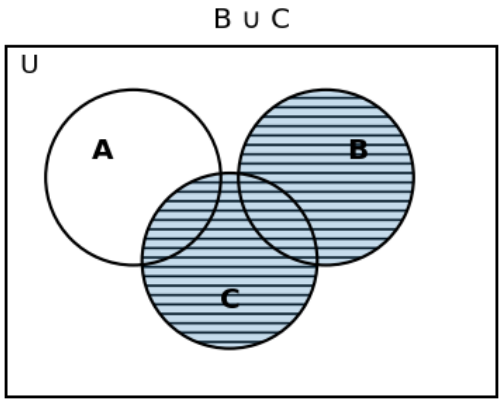
Figure (i)

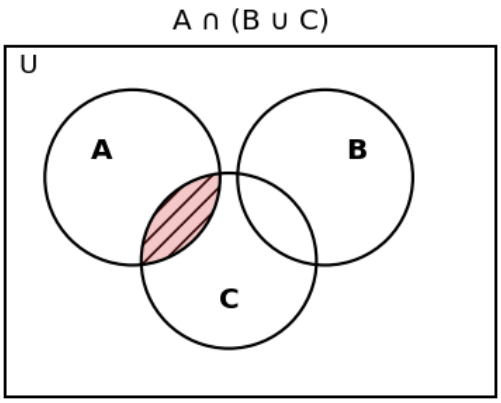
a) $A \cup (B \cap C)$ (first shade $B \cap C$, then the answer)



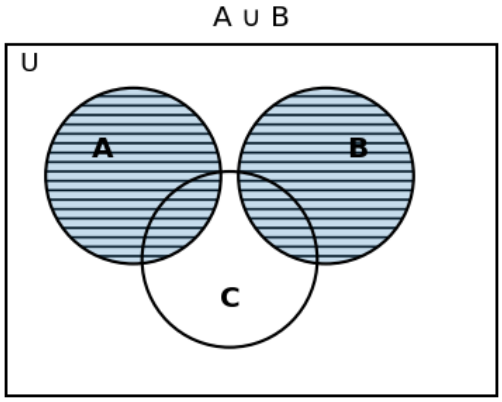


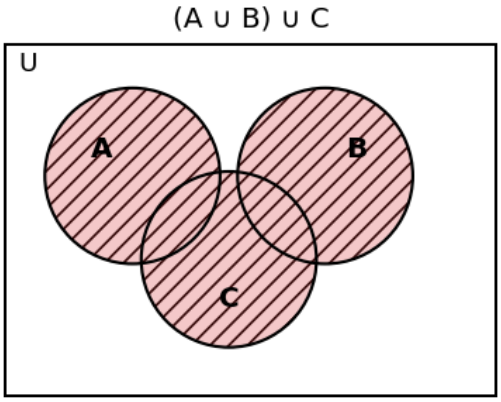
b) $A \cap (B \cup C)$ (first shade $B \cup C$, then the answer)



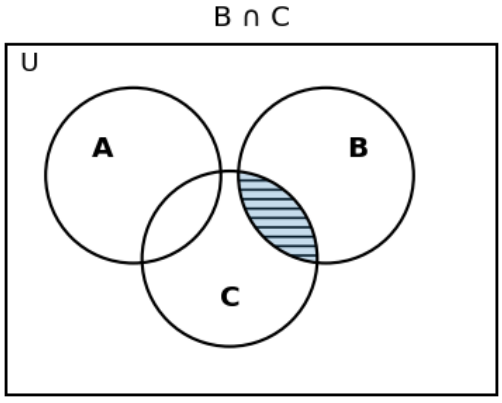


c) $(A \cup B) \cup C$ (first shade $A \cup B$, then the answer)





d) $A \cap (B \cap C)$ (first shade $B \cap C$, then the answer)



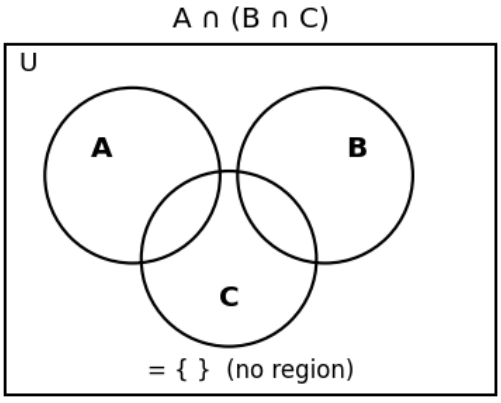
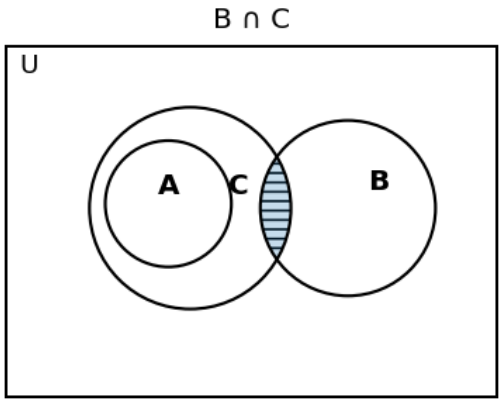
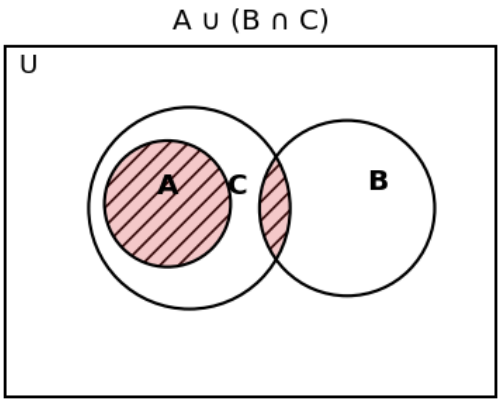


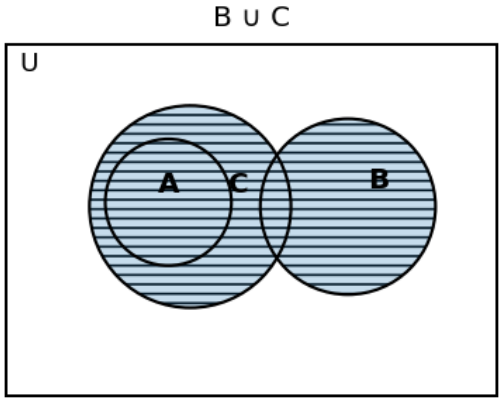
Figure (ii)

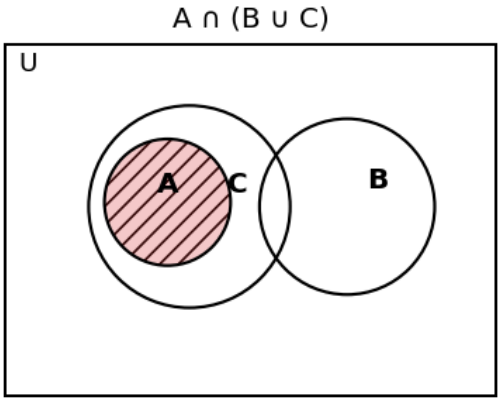
a) $A \cup (B \cap C)$ (first shade $B \cap C$, then the answer)



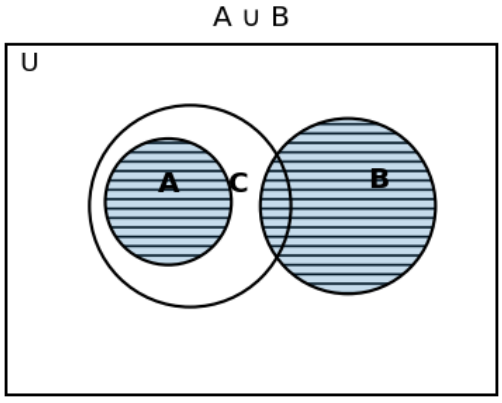


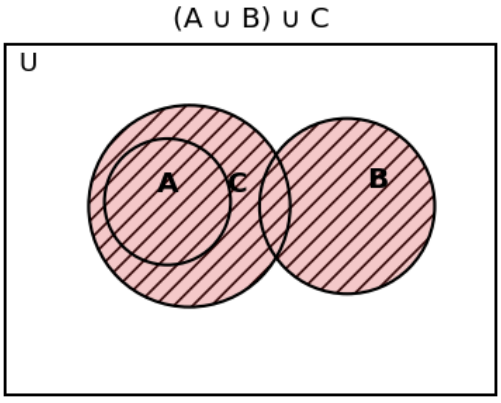
b) $A \cap (B \cup C)$ (first shade $B \cup C$, then the answer)



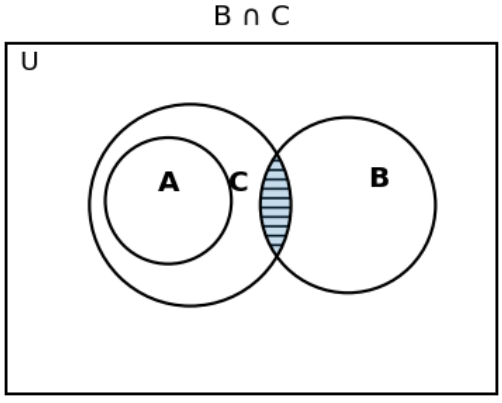


c) $(A \cup B) \cup C$ (first shade $A \cup B$, then the answer)





d) $A \cap (B \cap C)$ (first shade $B \cap C$, then the answer)



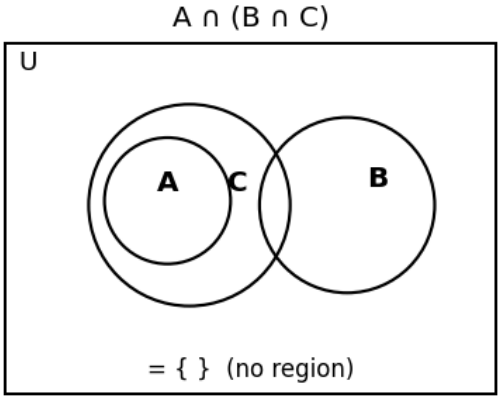
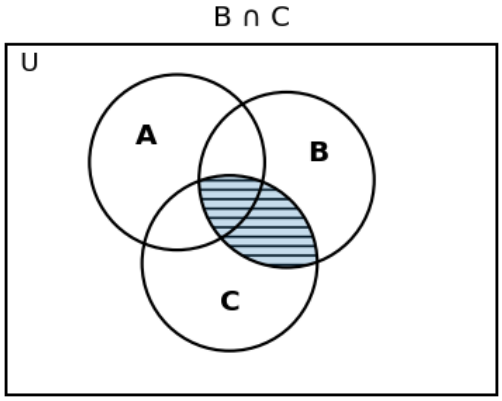
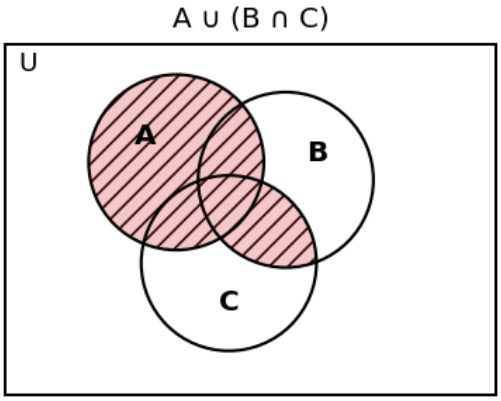


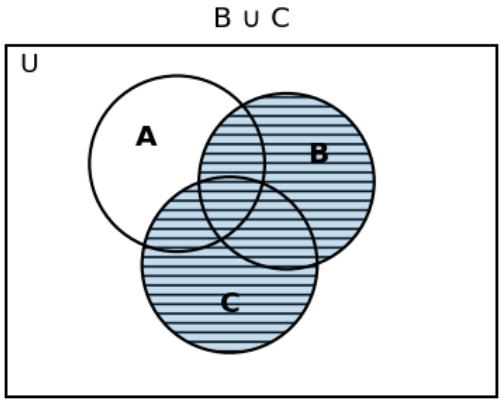
Figure (iii)

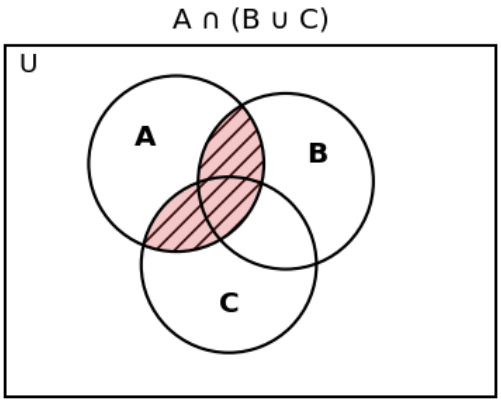
a) $A \cup (B \cap C)$ (first shade $B \cap C$, then the answer)



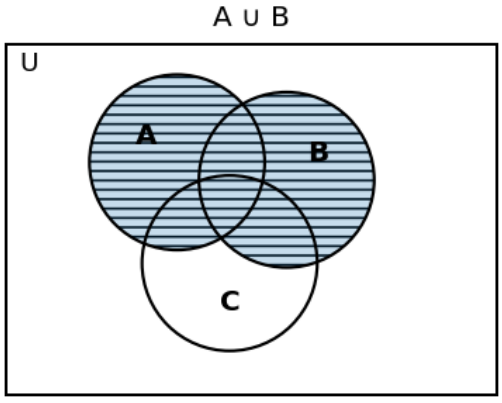


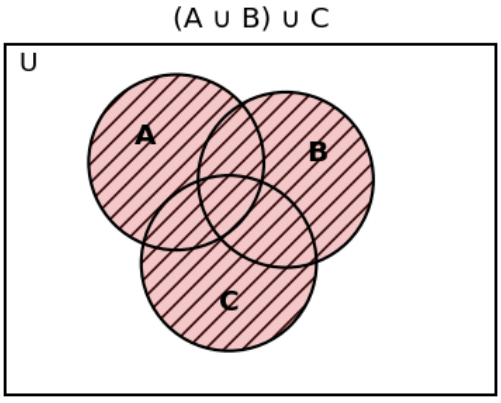
b) $A \cap (B \cup C)$ (first shade $B \cup C$, then the answer)





c) $(A \cup B) \cup C$ (first shade $A \cup B$, then the answer)





d) $A \cap (B \cap C)$ (first shade $B \cap C$, then the answer)

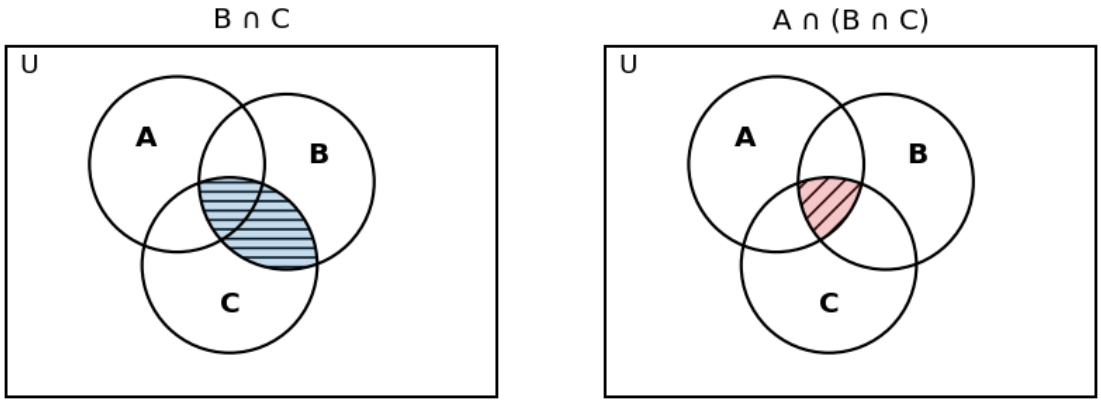
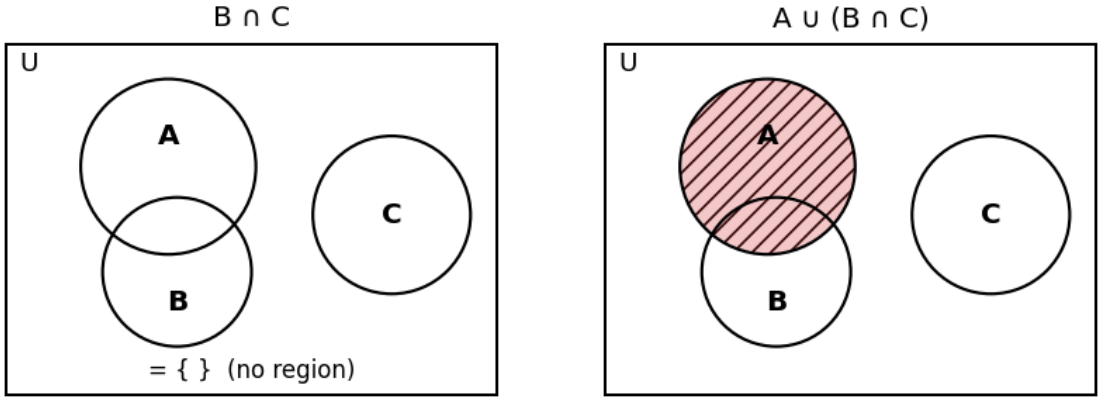
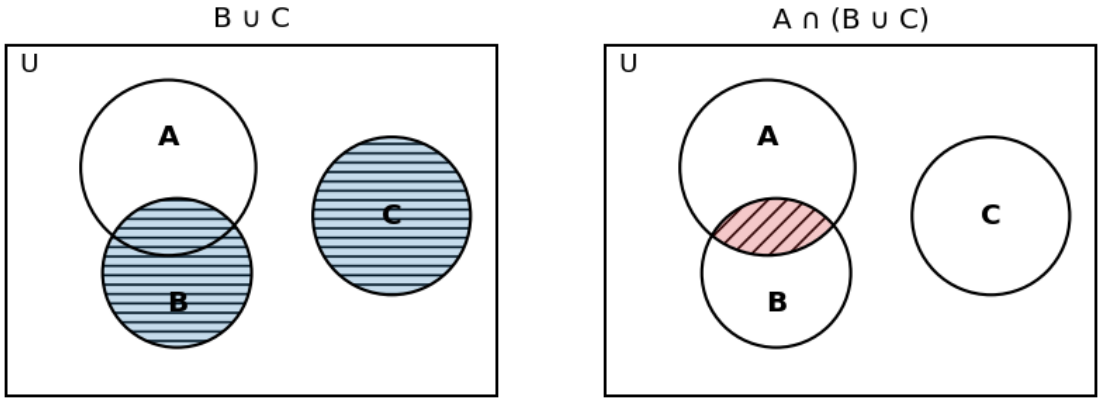


Figure (iv)

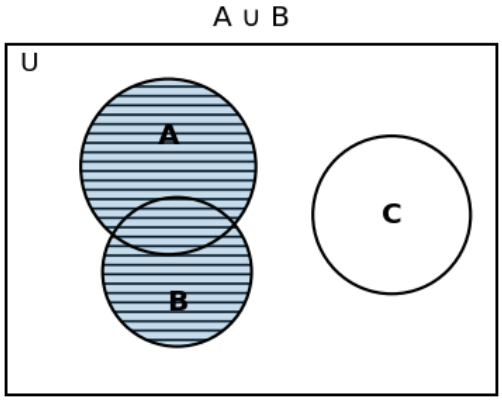
a) $A \cup (B \cap C)$ (first shade $B \cap C$, then the answer)

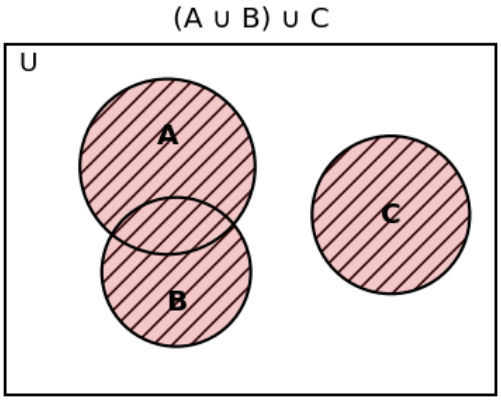


b) $A \cap (B \cup C)$ (first shade $B \cup C$, then the answer)

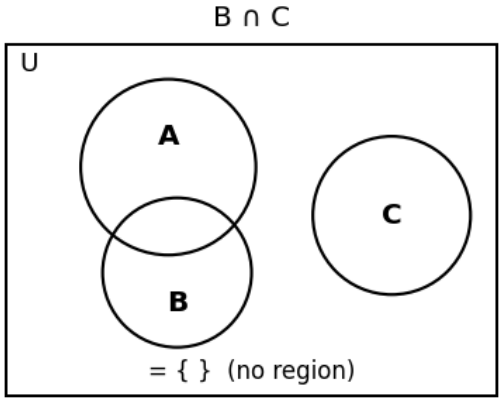


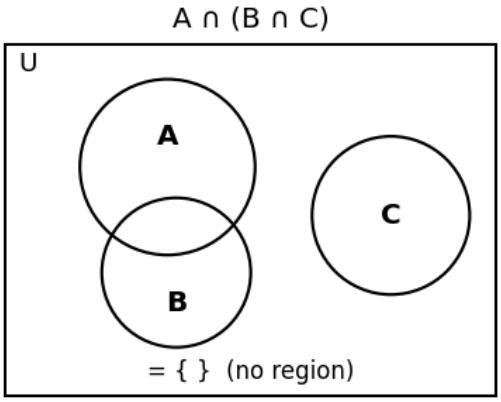
c) $(A \cup B) \cup C$ (first shade $A \cup B$, then the answer)





d) $A \cap (B \cap C)$ (first shade $B \cap C$, then the answer)



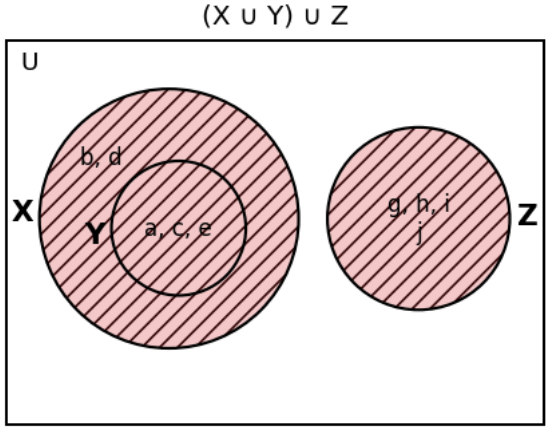


Question 2

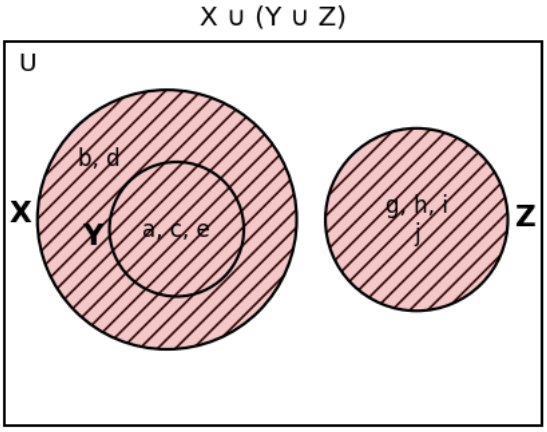
If $X = \{a, b, c, d, e\}$, $Y = \{a, c, e\}$, $Z = \{g, h, i, j\}$, then find the following using Venn diagram.

Here $Y \subseteq X$ (every element of Y is in X) and Z has no common element with X or Y.

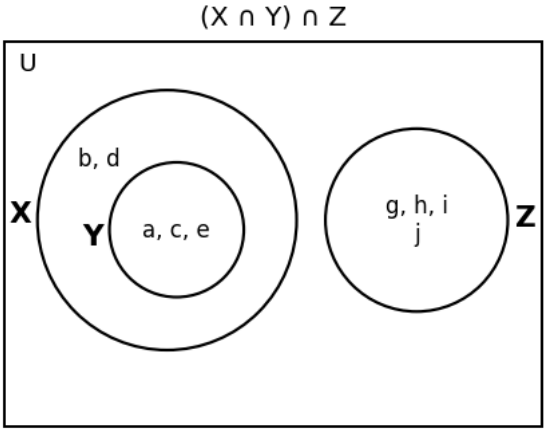
(i) $(X \cup Y) \cup Z = \{a, b, c, d, e, g, h, i, j\}$



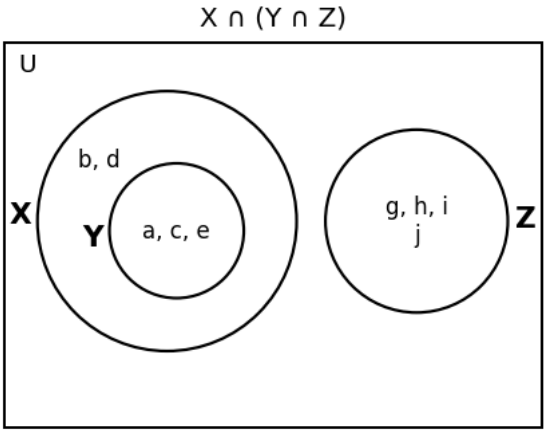
(ii) $X \cup (Y \cup Z) = \{a, b, c, d, e, g, h, i, j\}$



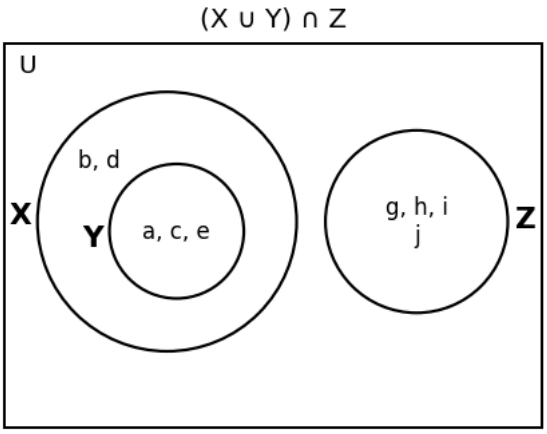
(iii) $(X \cap Y) \cap Z = \{\ \} = \phi$



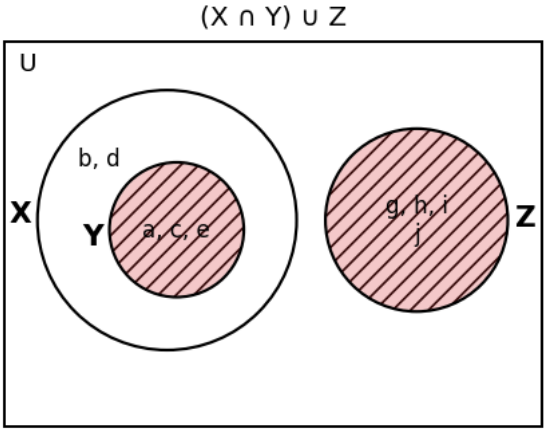
(iv) $X \cap (Y \cap Z) = \{\ \} = \phi$



(v) $(X \cup Y) \cap Z = \{\ \} = \phi$



(vi) $(X \cap Y) \cup Z = \{a, c, e, g, h, i, j\}$



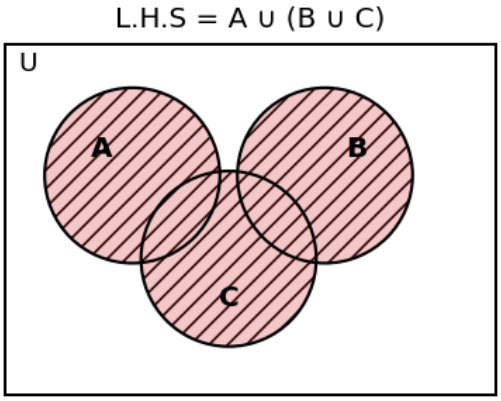
Question 3

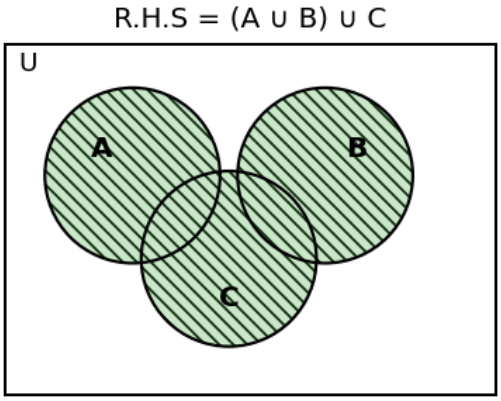
Verify the associative law of union and intersection by using the diagrams of Question 1.

(a) $A \cup (B \cup C) = (A \cup B) \cup C$ **(b)** $A \cap (B \cap C) = (A \cap B) \cap C$

Figure (i)

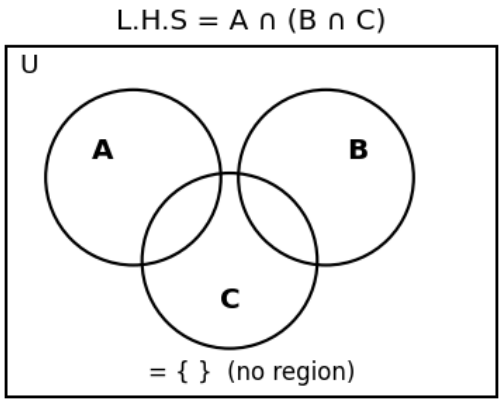
(a)

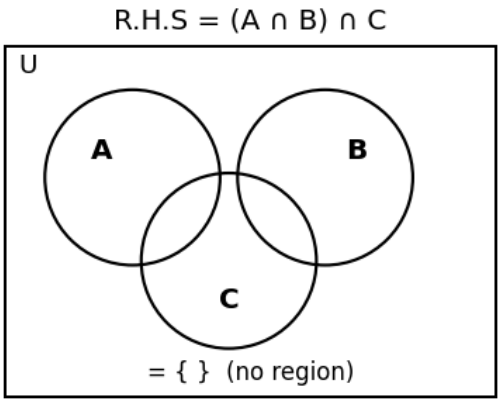




Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(b)

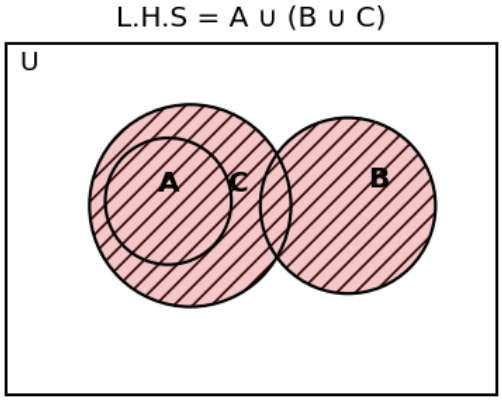


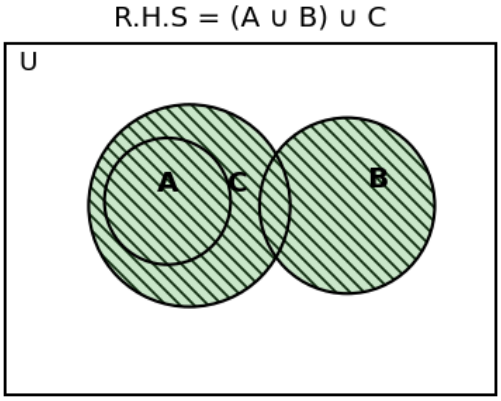


Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

Figure (ii)

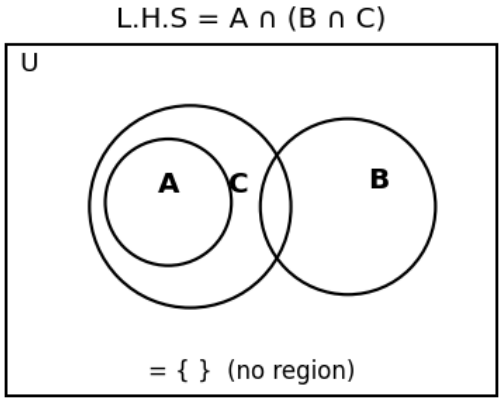
(a)

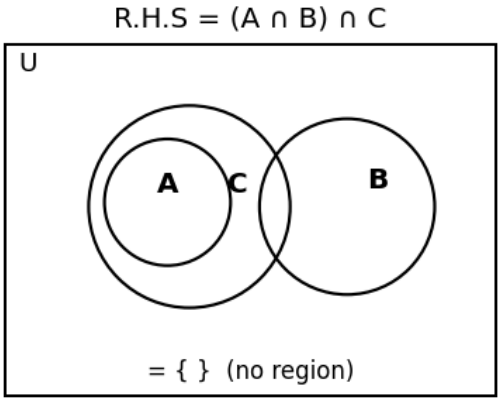




Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(b)

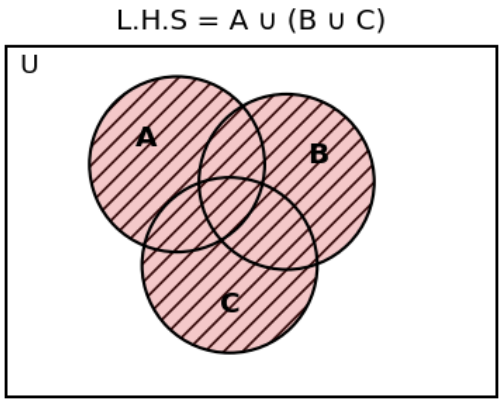


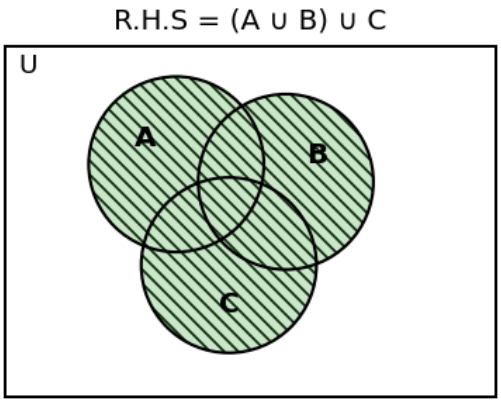


Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

Figure (iii)

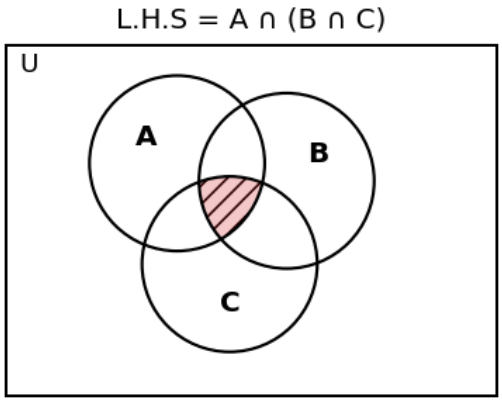
(a)

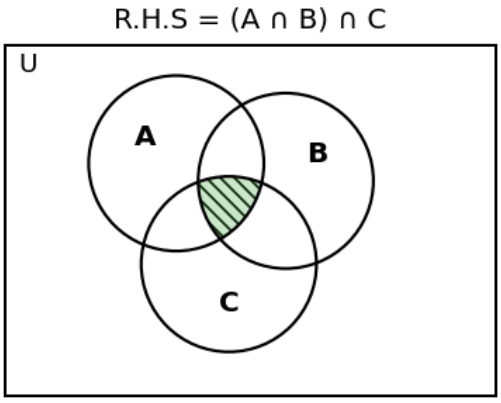




Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(b)

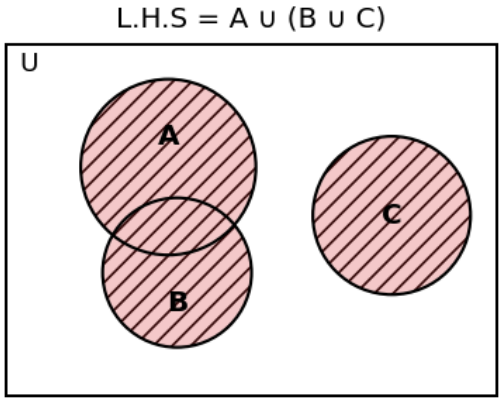


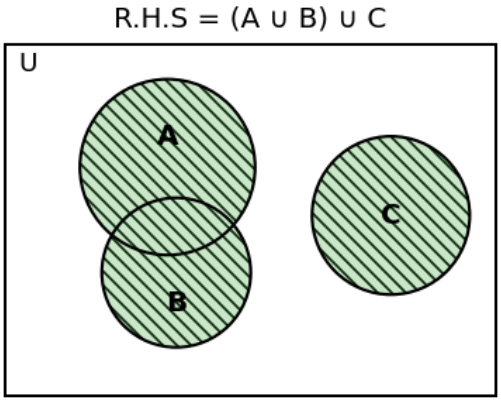


Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

Figure (iv)

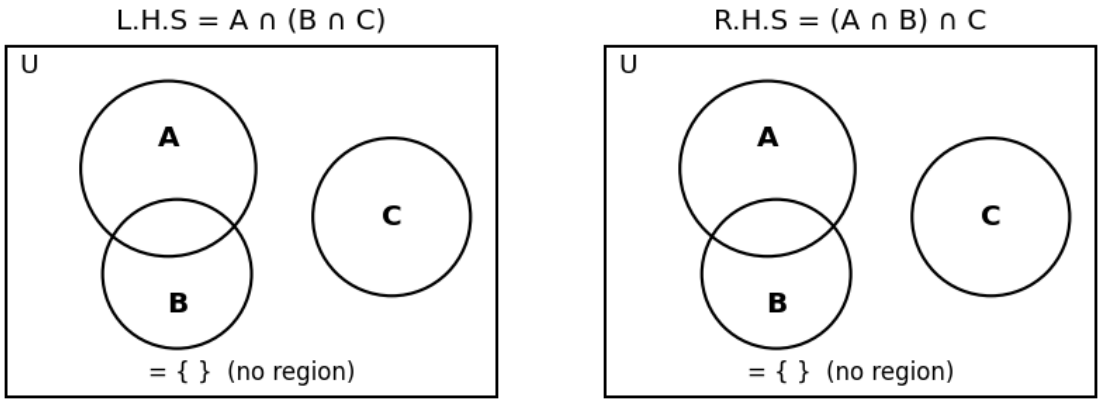
(a)





Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(b)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

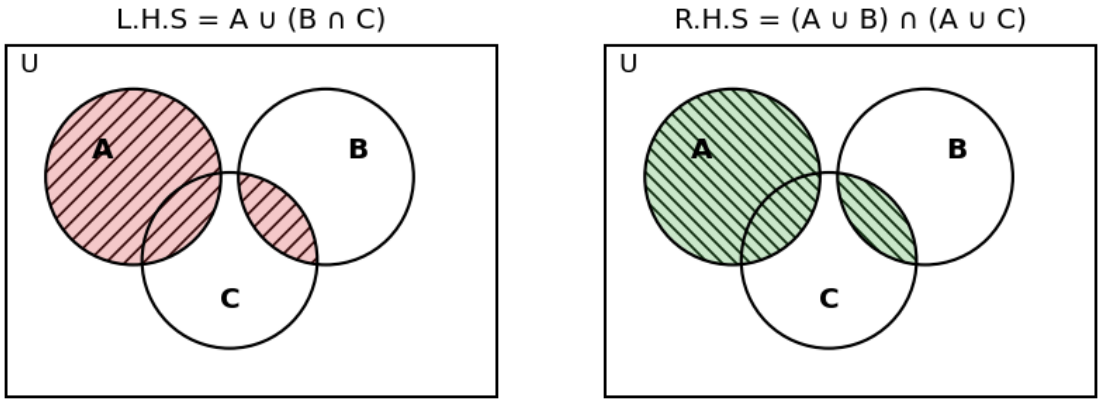
Question 4

Verify (i) distributive property of union over intersection (ii) distributive property of intersection over union, by using the diagrams of Question 1.

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ **(ii)** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

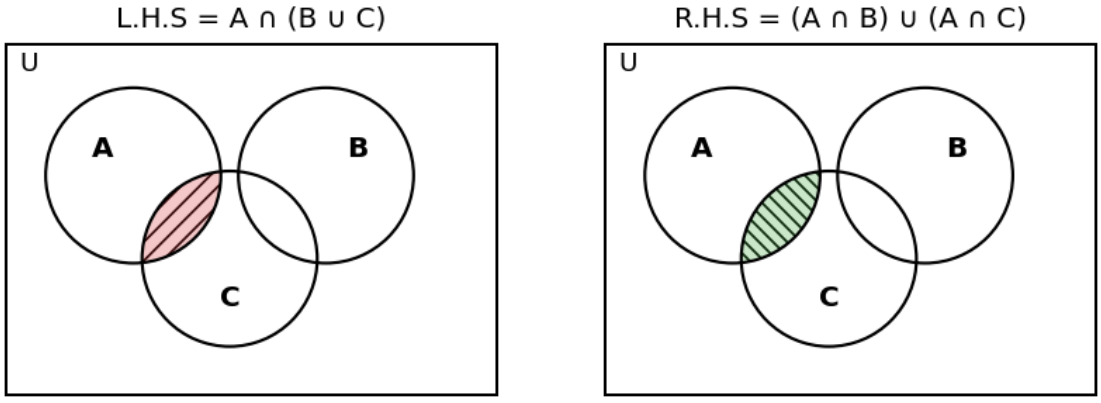
Figure (i)

(i)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

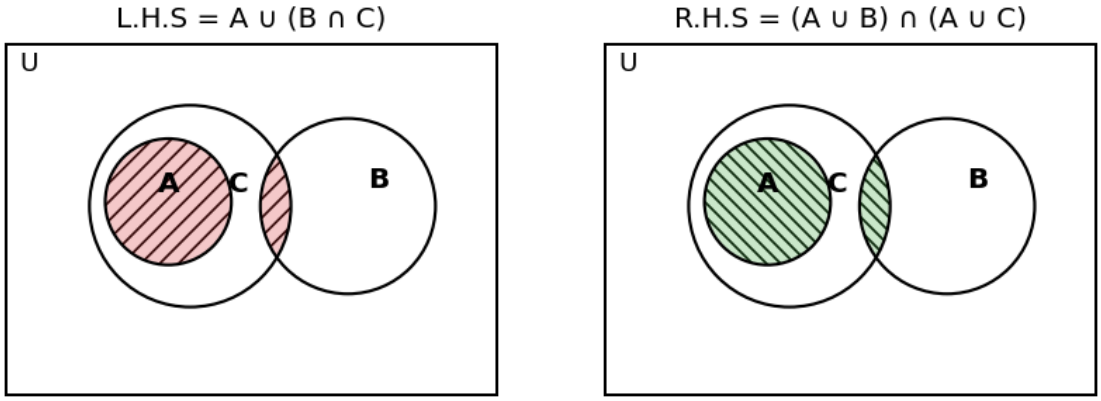
(ii)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

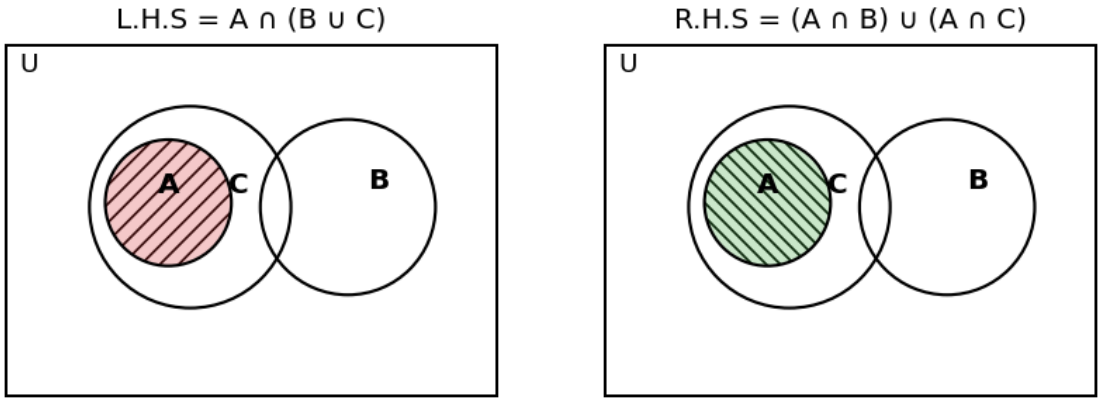
Figure (ii)

(i)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

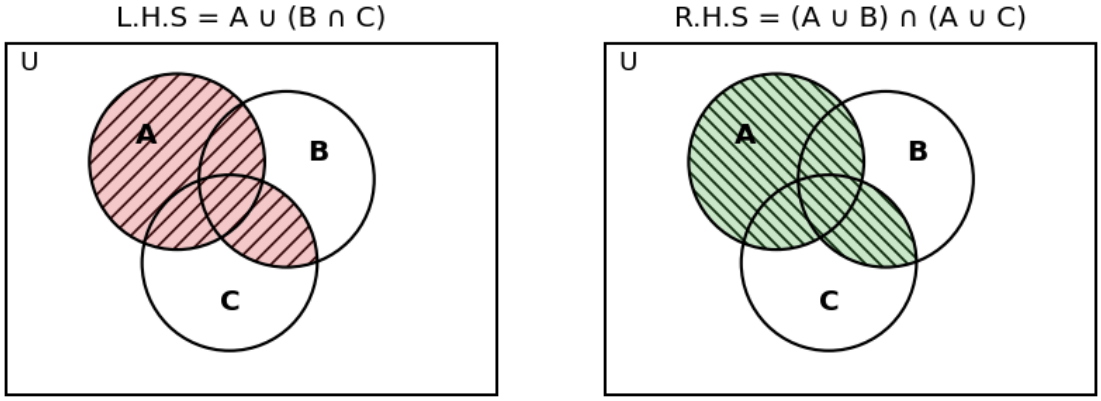
(ii)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

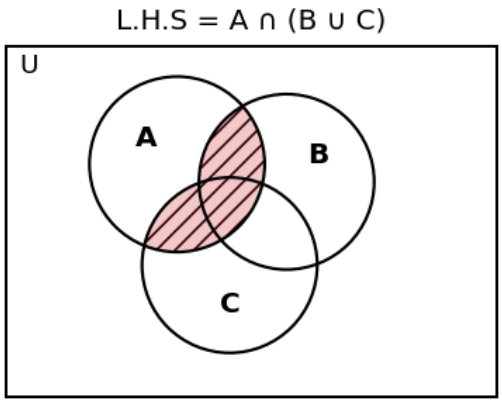
Figure (iii)

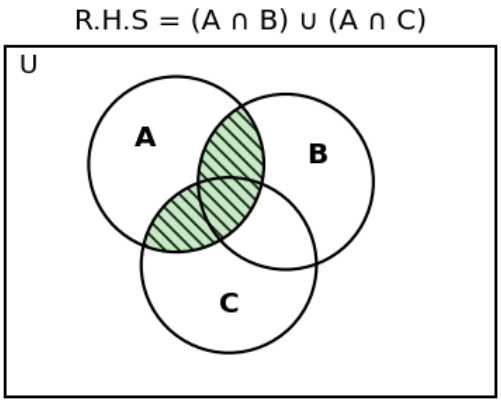
(i)



Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(ii)

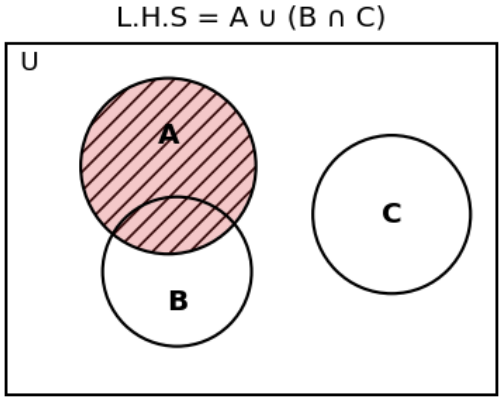


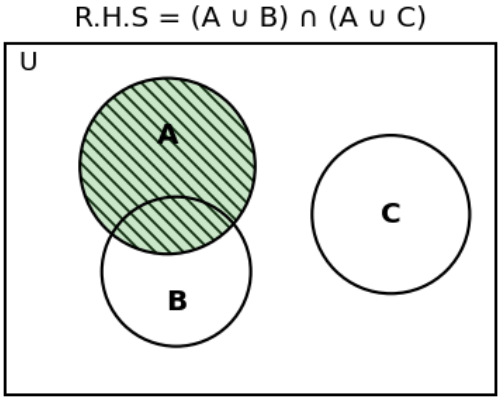


Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

Figure (iv)

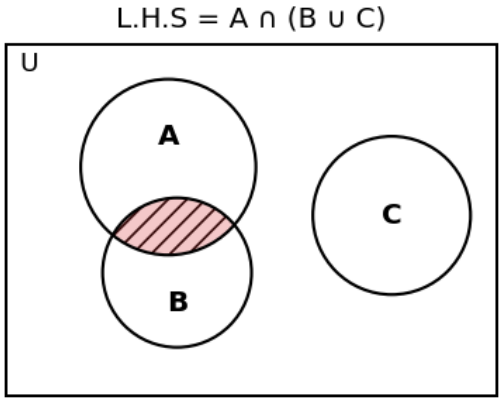
(i)

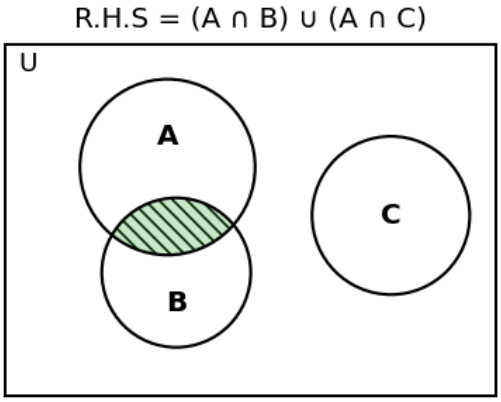




Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

(ii)





Same region is shaded in both diagrams, so **L.H.S = R.H.S**.

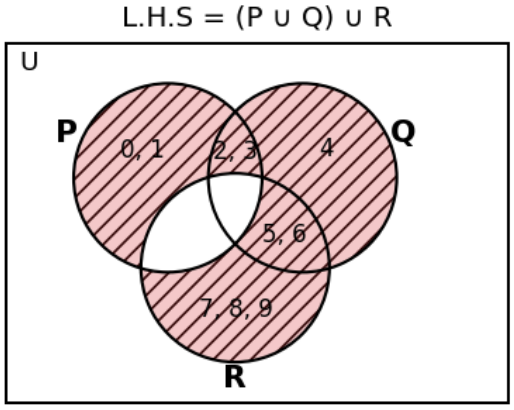
Question 5

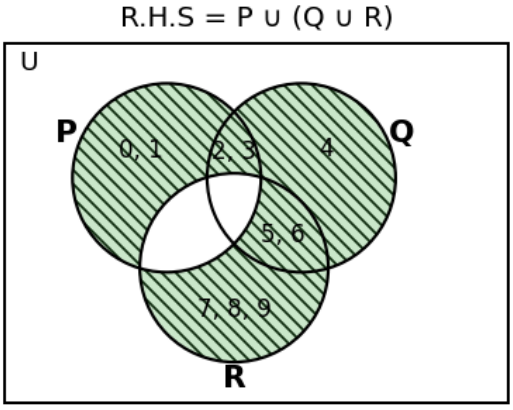
Prove by using Venn diagram: **(a)** $(P \cup Q) \cup R = P \cup (Q \cup R)$ **(b)** $(P \cap Q) \cap R = P \cap (Q \cap R)$

(i) $P = \{0, 1, 2, 3\},\ Q = \{2, 3, 4, 5, 6\},\ R = \{5, 6, 7, 8, 9\}$

(a) $P \cup Q = \{0, 1, 2, 3, 4, 5, 6\}, \quad (P \cup Q) \cup R = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$Q \cup R = \{2, 3, 4, 5, 6, 7, 8, 9\}, \quad P \cup (Q \cup R) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

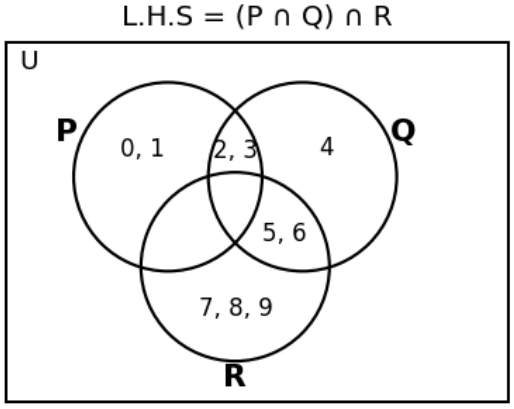


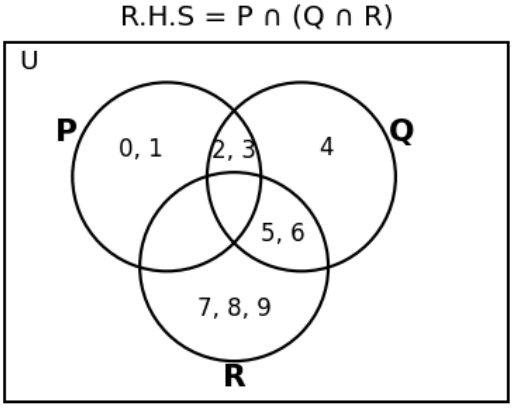


L.H.S = R.H.S, hence proved.

(b) $P \cap Q = \{2, 3\}, \quad (P \cap Q) \cap R = \{\ \}$

$Q \cap R = \{5, 6\}, \quad P \cap (Q \cap R) = \{\ \}$



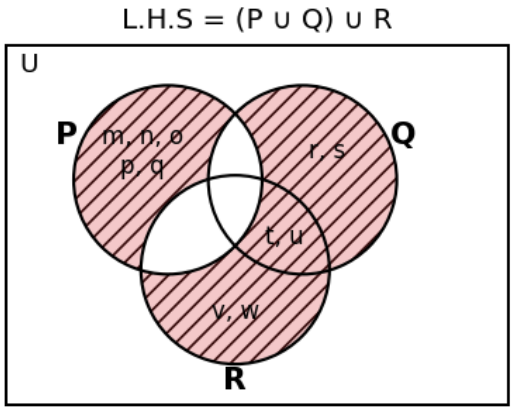


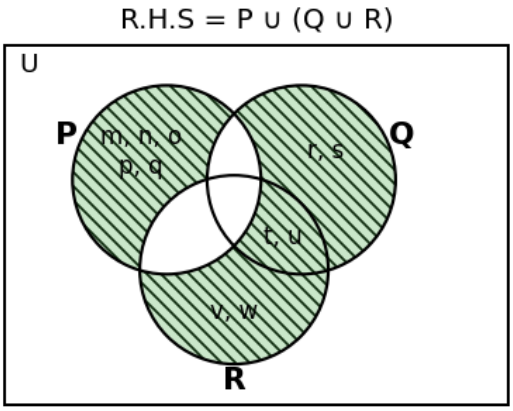
L.H.S = R.H.S (both are empty sets ϕ), hence proved.

(ii) $P = \{m, n, o, p, q\}, \ Q = \{r, s, t, u\}, \ R = \{t, u, v, w\}$

(a) $P \cup Q = \{m, n, o, p, q, r, s, t, u\}, \quad (P \cup Q) \cup R = \{m, n, o, p, q, r, s, t, u, v, w\}$

$Q \cup R = \{r, s, t, u, v, w\}, \quad P \cup (Q \cup R) = \{m, n, o, p, q, r, s, t, u, v, w\}$

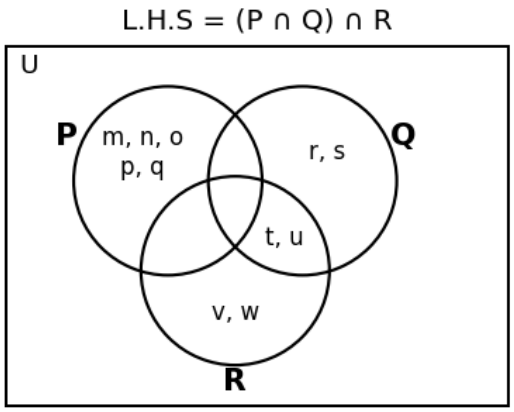


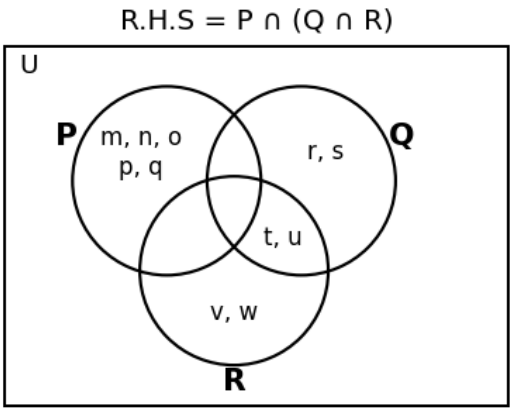


L.H.S = R.H.S, hence proved.

(b) $P \cap Q = \{\ \}, \quad (P \cap Q) \cap R = \{\ \}$

$Q \cap R = \{t, u\}, \quad P \cap (Q \cap R) = \{\ \}$





L.H.S = R.H.S (both are empty sets ϕ), hence proved.

Question 6

Verify $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$ using Venn diagram for the following sets.

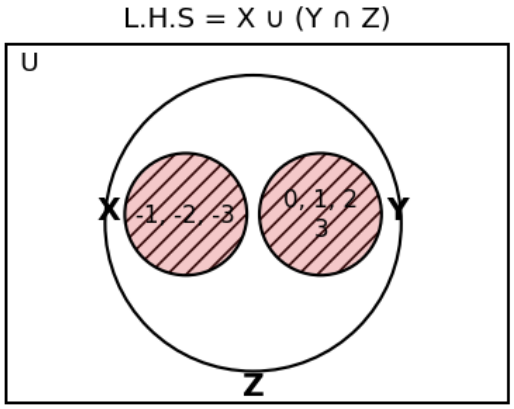
$$X = \{-1, -2, -3\}, \qquad Y = \{0, 1, 2, 3\}, \qquad Z = \{0, \pm 1, \pm 2, \pm 3\}$$

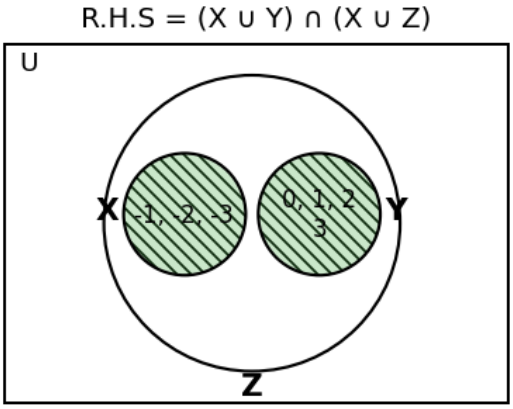
L.H.S: $Y \cap Z = \{0, 1, 2, 3\}$

$$X \cup (Y \cap Z) = \{-3, -2, -1, 0, 1, 2, 3\}$$

R.H.S: $X \cup Y = \{-3, -2, -1, 0, 1, 2, 3\}, \quad X \cup Z = \{-3, -2, -1, 0, 1, 2, 3\}$

$$(X \cup Y) \cap (X \cup Z) = \{-3, -2, -1, 0, 1, 2, 3\}$$





L.H.S = R.H.S, hence verified.

Question 7

Verify $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$

X = set of first three vowels, Y = set of letters of the word "energy", Z = set of letters of the word "algebra".

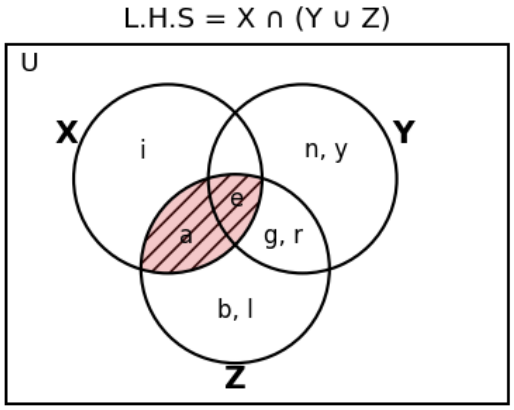
$$X = \{a, e, i\}, \qquad Y = \{e, g, n, r, y\}, \qquad Z = \{a, b, e, g, l, r\}$$

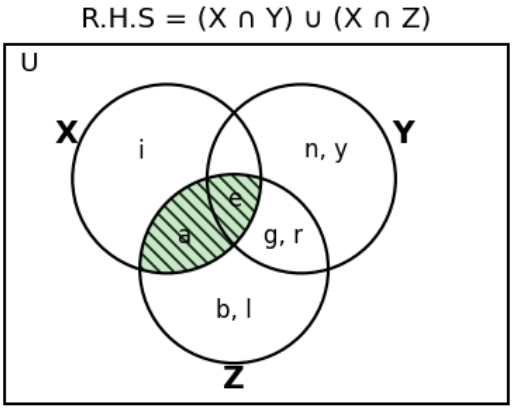
L.H.S: $Y \cup Z = \{a, b, e, g, l, n, r, y\}$

$$X \cap (Y \cup Z) = \{a, e\}$$

R.H.S: $X \cap Y = \{e\}, \quad X \cap Z = \{a, e\}$

$$(X \cap Y) \cup (X \cap Z) = \{a, e\}$$





L.H.S = R.H.S, hence verified.